



Aim

Explain how yeasts and other flavour-affecting factors can be managed to impact the final wine product.

INTRODUCTION

Good wine is certainly more than fermented grape juice. Complex flavours and aromas that are vital to the finished product need to be extracted from the grapes and developed through the winemaking process by intelligent and informed monitoring and managing each step in the process.

Characteristics of grapes used to make wine can vary a lot. Different varieties have (often) subtle differences in aroma and taste; and the conditions of harvest, and treatment after harvest can also cause differences in taste and aroma. Your choice of grapes, and how they are used can have a major impact.

Factors in the fruit may include:

Harvest stage – later picked grapes are sweeter and used to make sweeter fruitier wines.

Presence of fungus – late picked wines infected with botrytis rot or Noble rot. If the conditions are cool and moderately dry, then the fungus will live on the fruits and extract moisture without causing it to collapse. The fruits shrivel, the sugar, acid, flavour and aroma constituents become highly concentrated producing a sweeter botrytis wine.

- Grape skins – their characteristics and what is done with them
- Grape Seeds - flavour

Factors beyond the fruit may include:

- Yeast
- Acidity
- Sugar levels
- Fungus and other microorganisms
- Additives
- Perception of astringency

PREPARING TO FERMENT

- Addition of Sugar: When this comes into contact with yeast it turns from a white powder into a fizzy gas (carbon dioxide), and a simple molecule called ethanol (alcohol). There is much debate as to the quantity of sugar added. It must be a minimum of 10% alcohol by volume which will produce the minimum percentage of alcohol to preserve the wine. On the other hand, too much sugar will produce undrinkable syrup. The precise measurement of the amount of sugar is referred to as gravity and is measured using a hydrometer.
- Wine is usually started with a gravity of between 1090 and 1100 (though some home winemakers start with a kilo of sugar a gallon of must, and then add up to another 8 oz. later on). The first method gives more control over proceedings. More ?

sugar is added to sweet wines than dry ones, and alcohol will usually end up at around 15%.

- Tannins may be added. This may be in tablet or liquid form.
- Pectic enzyme: It may be desirable to add some to get rid of haze induced by pectin. It is available as a powder or a liquid.
- Acid is added: If the finished wine is too acidic it will taste sharp. If it is too alkaline it will taste bland. Winemakers aim to be somewhere between the two extremes. The universal measure of acidity is parts per 100,000 of sulfuric acid (ppt).

Wine ranges from 3.0 ppt for a light dry wine to 4.5 ppt for a sweet full-bodied wine. Winemakers attempt to get the ppt within 0.1 ppt of where they want it by adding acid or calcium carbonate, to raise acidity or alkalinity, respectively. There are a number of different acids available for this purpose: Citric, Malic, Tartaric, Succinic and Tannic

- Yeast is added: Yeasts should be stored at temperatures that do not fluctuate too much, and they need to be fed with yeast nutrients or sugar. They also prefer a slightly acidic environment. There are many different types of yeast to choose from including standard wine yeasts and more specialist ones.

The whole mixture is now stirred and left at a temperature above 15°C to allow it to ferment. The fermenting must is usually strained after a few days. Homemade wines are put in demijohns at this point and fitted with an airlock watch.

- The mixture is left for weeks to months until the fermentation process stops. This can be determined by:
- A specific gravity of 1005 or less
- No bubbles in the airlock
- The wine begins to clear from the top
- The wine is now racked (siphoned) leaving the gunk (lees) behind.
- Clearing and finishing is the final stage. The aim is to now turn the siphoned wine into crystal clear wine that is bottled and labelled. Stabiliser and crushed Campden tablets may be added; after 24 hours no further fermentation will take place, and the wine will be stabilised. Wine finings can be added to remove any residue within the next week

The wine is then racked again, tasted to adjust acidity/alkalinity levels if needed, filtered, and then poured into bottles, corked and left to mature

YEASTS AFFECT FLAVOUR

There are different types of yeasts that can be used in the wine making process. Different yeasts have different flavours in themselves, but they can also affect the flavours that come from the grapes during the fermentation process and in that way, ultimately affect the flavour of the wine.

There are a few important things to remember about yeasts:

1. They grow in an environment that excludes air
2. Unlike many other microorganisms, yeasts tolerate high levels of sulphur dioxide
3. Yeasts tolerate pH as low as 2.8 (lower than other microorganisms)
4. Wine cellars that have been used for a long time tend to have high levels of yeasts occurring naturally on the air.

Some types of yeast and other microorganisms occur naturally in the air and on the surfaces of tools and equipment used in a winery. These naturally occurring microorganisms are not all good, and they can affect fermentation in a negative way. These microorganisms can include a bacterium that converts alcohol into acetic acid (i.e., *Acetobacter* species), and in doing so can convert wine into vinegar. *Acetobacter* and many other undesirable microorganisms will not multiply anaerobically; so, wine will not be tainted and turned to vinegar so long as air is excluded from the fermenting wine.

There are 2 types of yeasts used in wine making:

1. Wild or naturally occurring yeasts
2. Cultured yeasts

Common Yeast Strains

Whilst *Saccharomyces cerevisiae* is widely used, other *Saccharomyces* strains that are used in winemaking include:

- *Saccharomyces bayanus*
- *Saccharomyces beticus*
- *Saccharomyces fermentati*

- *Saccharomyces paradoxus*
- *Saccharomyces pastorianus*
- *Saccharomyces uvarum*

Different strains can impart some difference to the flavour of a wine but not as much as different grape varieties do. A good wine is not just dependent on the yeast. Common yeast is agar, liquid, or dry.

Cultured Yeasts

Some (not all) cultured yeasts are useful for fermentation of wine. The most useful is the species *Saccharomyces cerevisiae*.

There are differences between different varieties (strains) of this species. It is important to use an appropriate type for a particular result. Microbiologists are involved in collecting, culturing and providing appropriate species for use in producing the desired wine. Winemakers use small quantities of known cultured yeast to make a “starter culture” and grow a larger quantity of yeast to use in the wine-making process.

Wild Yeasts

Wild or naturally occurring yeasts may be used to start the fermentation processing a less “controlled” way. Several different yeast and other microorganism species will naturally cause fermentation to begin.

The involvement of non-wine yeasts can diminish the flavour/quality of wines that are fermented this way.

Once alcohol level reaches 4 to 6%, growth of non-wine yeast organisms will be inhibited naturally and wine yeasts will take over at that stage. Adding sulfur dioxide is a way of further inhibiting the non-wine yeasts, hence improving the flavour.

Wild Yeast Types

Although wild yeast refers to those found on the grapes it can also refer to yeasts which come into contact with the must from winemaking equipment. In wineries these are usually strains of *S. cerevisiae* often occurring as remnants of previous vintages. These yeasts are also called ambient or indigenous yeasts as opposed to inoculated or cultured ones. There is debate as to how they contribute to wines. Some argue that they give wines from particular wineries their ‘terroir’ or unique flavour. However, they are less predictable than inoculated strains and can induce off-flavours or aromas and an overly acidic taste. They are also not as vigorous meaning that not all sugar is converted to alcohol, so fermentation remains incomplete. This is known as ‘stuck’.

Some wild yeast types include:

- *Aureobasidium*
- *Brettanomyces*
- *Candida*
- *Kloeckera*
- *Saccharomycodes*
- *Schizosaccharomyces*
- *Zygosaccharomyces*

Of the above, *Kloeckera* is regarded as the most common one found in vineyards. Some species of *Kloeckera* actually kill off less vigorous strains of *Saccharomyces cerevisiae* through the production of ethyl acetate and acetic acid. *Brettanomyces* is responsible for off-flavour in wine producing a distinctive ‘barnyard’ like smell. Some winemakers consider that a small amount of this adds a particular quality to some wines such as pinot noir. Others are less forgiving and regard ‘Bretty’ wines as faulty.

Keeping Yeasts Happy

Yeasts need carbohydrates (e.g., Sugars in grapes) to grow, but not carbohydrates. They also require nitrogen plus a range of trace elements, minerals, and vitamins. Grape juice is ideal for making wine because it contains all of these things.

Some components (e.g., Tannins) found in grapes are not used by yeast but can still impact the appearance and flavour of the wine.

MANAGING YEASTS

The success of wine making depends heavily upon managing the presence and activity of yeasts. Desirable yeasts need to be encouraged, and undesirable yeasts and competing microorganisms need to be suppressed if not eliminated.

To do this a winemaker focuses on three things:

1. Strict hygiene
2. Excluding air from the fermenting “must” and wine.
3. Using sulphur dioxide

TEMPERATURE

Wines ferment faster at higher temperatures, but the rate of fermentation can affect flavour and ultimate quality. At temperatures over 20°C wine might only take 2 weeks to ferment.

White wine is generally a better quality when fermented at lower temperatures and is more commonly fermented at a temperature between 10 and 15°C. Red wines can be fermented safely at up to 25°C.

As wine ferments though, the fermentation process can generate heat, and if a large quantity of wine is fermented, overheating can become an issue.

MEASUREMENTS

Winemakers use a range of unique measurements to describe their products, including:

- Degrees Brix
- Degrees Baume
- Degrees Balling

DETERMINING ALCOHOL CONTENT IN WINE

Winemakers can determine the potential alcohol concentration if all this sugar is converted to alcohol. They can also tell how much sugar they will need to add to the juice to produce alcohol of the desired concentration.

Ebullioscope

This method is based on the variation between of boiling point of water and alcohol-containing solution. The Dujardin Salleron types ebullioscope or ebulliometer is best known.

Method

Rinse the boiling chamber of ebullioscope carefully with distilled water.

1. Add approximately 25mL of distilled water to the boiling chamber and insert the thermometer.
2. Screw the condenser into position and apply the heat.
3. Bring the water to boil and continue until the temperature remains constant.
4. Record the temperature. Set the slide rule so that the 0.00 mark of alcohol is opposite this temperature.
5. Empty the chamber and rinse with the wine sample.
6. Pipette 50ml of the wine into the boiling chamber. Reinsert the thermometer and screw the condenser filled with cold water into position and apply the heat.
7. Observe the temperature as the wine boils. Note the temperature when the thermometer remains static and switch off the heat. Read off the percentage of alcohol in the wine sample opposite the observed temperature on the scale marked 'wine'.

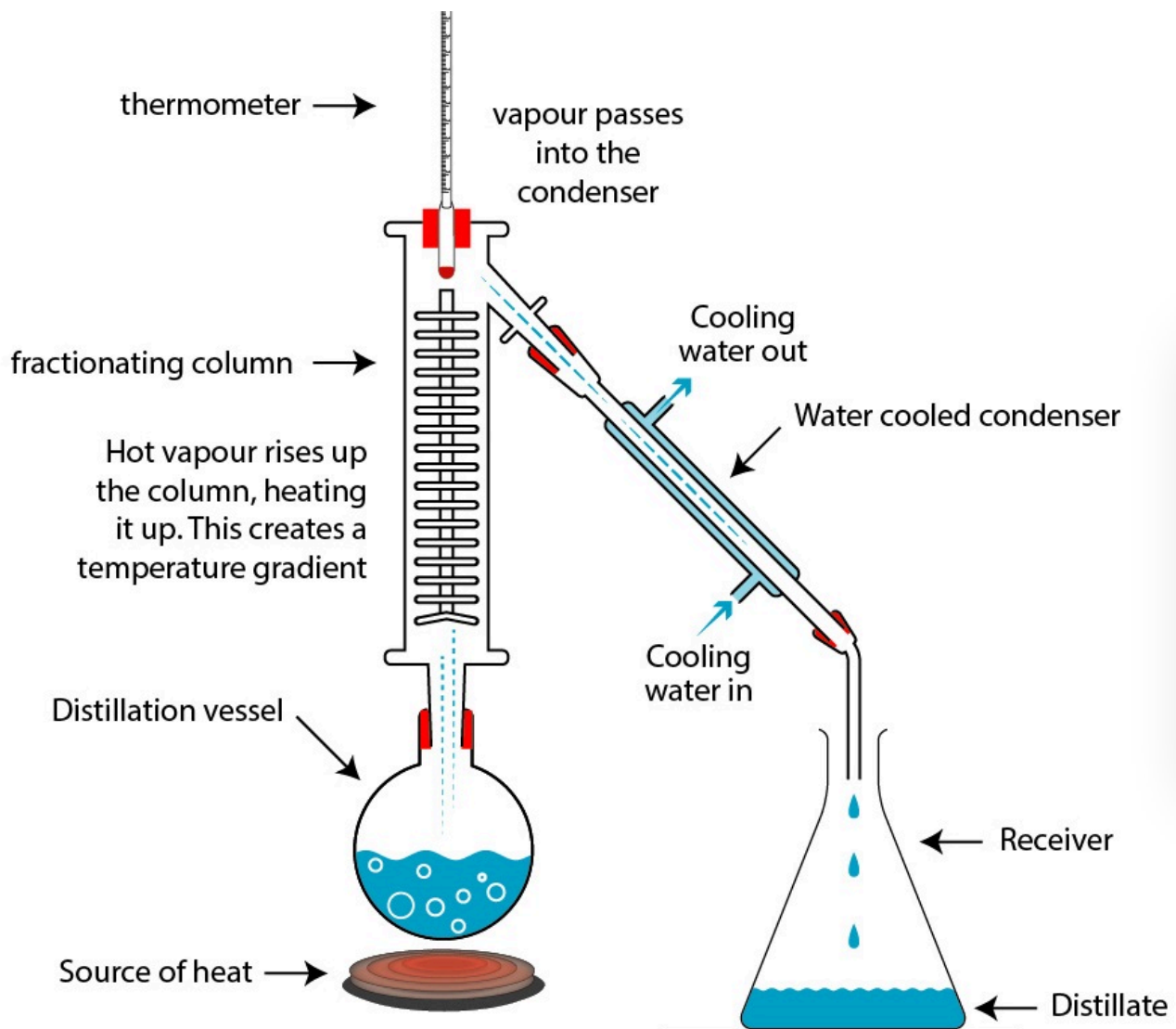
Note: Sweet wine sample must be diluted to 5% sugar concentration before boiling, and the result is multiplied by the dilution factor.

Distillation

Distillation of wine sample distinguishes the volatile from the non-volatile components.

Method

1. Measure 250ml of wine sample and transfer into 1 litre boiling flask containing 300ml water and few glass beads.
2. Connect the condenser and apply heat.
3. Distil about 200ml of the sample into a volumetric flask containing 20ml water and ensure that the condenser's outlet is submerged throughout the distillation.
4. Makeup the distillate to 250ml with water and mix well.
5. Float alcohol or specific gravity hydrometer in the distillate and record the reading. Make sure the temperature of the sample is the same at which the hydrometer is calibrated. Temperature can be adjusted in a water bath if necessary.
6. Use the conversion table if the reading is done with a hydrometer designed for measuring specific gravity.



DETERMINING MALOLACTIC FERMENTATION

Malo-lactic fermentation (MLF) is a natural process for adjusting acidity. It is carried out mostly in red wines and some sparkling and white wines. MLF lowers the acidity by converting malic acid to lactic acid. A paper chromatography process is used to assess the presence of malic acid and verify if malo-lactic fermentation has occurred.

Reagents and Equipments:

- Eluting Solvent: mix 60ml of n-butanol in 60ml of water, add 6.4ml of concentrated formic acid and 9ml of 1% bromocresol green indicator. Transfer into 250ml separating funnel fitted with Teflon tap. Mix thoroughly by shaking and leave overnight. The solvent will be clearly separated into two distinct layers. Discard the bottom layer. The top layer is then transferred into 2-litre beaker covered with a glass sheet. If complete separation of water from solvent is difficult, filter the upper layer through several fluted Whatman's No 1 filter paper.
- 5 % Malic acid solution: Dissolve 5g of L- malic acid in 50 ml water and make up the total content to 100ml with water. It will keep up for 4 weeks in the refrigerator.
- Chromatography paper: Use 25 X 25 cm sheets of Whatman's No 1 paper.

Method

1. Mark a pencil line at about 2cm intervals from the bottom of the chromatography sheet.
2. Place a small amount of malic acid solution on the first marked point about half a centimetre across using a capillary tube or micro-loop made of thin wire. If using a wire loop, clean it by washing in purified water, flaming, and drying before each application. If using a capillary tube, use a new tube for each sample.
3. Now place the drops of wine sample along the 2cm marked lines until all samples are represented.
4. Allow the spots to dry for a minimum of 15 minutes. Place the chromatography paper in a 2-litre beaker containing no more than 1cm depth of solvent. The pencil line and spots must be at the base above the solvent level. Cover the beaker with a glass sheet.
5. Allow it to stand for about 2-3 hours or until the solvent travels at least 15-20cm up the paper.
6. Remove the paper and hang it to dry.

Result

- Organic compounds move upward in a column-like fashion at different rates in each spot, and yellow-coloured spots on dried paper indicate the position of acids on a blue-green background. The colour intensity indicated the presence or absence of this acid.
- Observe the location of each spot, and classify the acids present in the tested wine sample by contrast with the position of the regular spots.
- This colour can fade over time and can be revived by holding a formic acid bottle's paper.
- In wine sample, the acids will appear in the following order from the base line: tartaric, malic, lactic, and succinic acids.

DETERMINING MICROBIAL STABILITY IN WINE

The activity of microorganisms such as yeast or bacteria in bottled wine can cause a serious problem. Hence correct use of filtration and sterile bottling is an integral step in wine production, especially with freshly bottled sweet wines or red wines which have been prevented from undergoing malo-lactic conversion

Method A - Agar Plating

Reagents and equipment

- Agar 1% - Make 1% solution of potato dextrose agar (P.D.A agar) in water
- Autoclave or large pressure cooker
- Petri dishes
- Microscope: A low magnification microscope (40 -100x) is suitable
- Micro-loop: a loop nichrome wire (e.g., wire from a heater coil) inserted in a wooden handle
- Source of flame: Burner, alcohol lamp or candle

Procedure

1. Autoclave petri dishes and 1% agar in a bottle (loosen lid) for 15 min at a pressure of 15psi.
2. Pour the agar onto the plates while hot, replace the lid and leave to set.
3. Sterilise the wire loop in an open flame and, by streaking over the agar surface, pass small quantities of the sample.

4. Now place the sealed plate in an incubator or sealed cupboard for 48 hours at about 25-30 °C.

5. Note the presence of any growth on the agar surface. A microscopic examination will assist in the identification of microorganism presence- some expert help may be needed.

Method B – Plate count technique

The advantage of this method is that it checks the purity and estimates the degree of contamination.

Procedure

1. Makeup dilution of each sample as shown below and add 1ml of each sample to separate sterile petri dishes.
2. Then pour some more quantity of warm sterilised agar medium on each plate and mix until agar sets by swirling petri dish.
3. Place the sealed plate in an incubator or sealed cupboard for 48 hours at about 25-30 °C and observe the appearance of growth.
4. The quantity of yeast or bacteria in 1ml sample = No. of colonies counted in the contaminated plate X dilution factor

Dilutions use:

- 1ml of the sample into 99ml of salt/water (1:100 ratio)
- 1ml of the diluted sample (a) into 99ml of salt/water (1:10 000 ratio)
- 1ml of the diluted sample (b) into 99ml of salt/water (1:1 000 000 ratio)
- Salt solution: Add salt at 5% to tap water and boil for 10 minutes. Do not use distilled or chlorinated water.

Note: With both the methods, to prevent condensation accumulation on the lid store, all the plates upside down after pouring agar.

DETERMINING CHEMICAL STABILITY IN WINE

Grape juice naturally contains a significant amount of tartaric acid throughout its ripening period. This acid is responsible for much of the wine's tart taste and contributes to the wine's durability and biological stability. Before bottling the wine should be checked for its basic stability. A stable wine should not be affected by moderate temperature change or vary unduly in chemical composition over time. An unstable wine will need to be treated by one of the methods described below.

Method

1. Pour equal quantities of filtered wine sample in two bottles (A and B)
2. Keep Bottle A at 40°C for 48 hours.
3. Keep Bottle B at -2°C for 48 hours.

A wine bottle that remains clear will be chemically stable, and the one that is not clear will have sediments or hazes of the following types:

- Crystals: The formation of crystals indicates the wine is tasteless, off-white and lacks cold stability. Restabilise by chilling at -4 °C for a minimum of four days and filter before bottling.
- Hazes: The formation of haze indicates either chemical or microbial instability. Chemical instabilities could be metal, phenol or protein hazes and can be determined as follows:
 - Metal hazes: Add 5ml of hydrochloric acid (HCl) to 45ml of wine sample. If haze disappears, it was due to the presence of copper or iron.
 - Phenol hazes: Mix equal part quantities of wine sample and absolute alcohol. If haze disappears, it can be caused by tannin or phenol-based pigments. Gelatin fining followed by filtration will overcome this problem.
 - Protein hazes: This haze appears during the heat test. The test involves including a haze wine sample at about 80°C for 8 hours to unfold the haze forming proteins and then cooling the wine at 20°C for 3 hours. Examine the sample by passing an intense light after filtration. If the light is visible passing through the wine sample, then the result is considered as fail.

Another option is to measure initial turbidity of the filtered sample using a turbidimeter. A turbidity reading of 2.0 NTU or less is recommended. Protein hazes are removed by fining with bentonite or similar agents, followed by filtration.

DETERMINING TOTAL ACIDITY

The organic acids in wine are extracted mainly from grapes. The acid content is determined by titrating a sample with a base such as sodium hydroxide solution to a phenolphthalein endpoint.

Reagents

- Potassium hydrogen phthalate: Dissolve 0.7g of potassium hydrogen phthalate in 50ml hot water. Make this solution fresh.
- Phenolphthalein indicator: Dissolve 1g of phenolphthalein in 50ml of pure ethyl alcohol and dilute to 100ml with water.
- 0.133N sodium hydroxide: Dissolve 22g of sodium hydroxide pellets in 500ml of water. Makeup to 4 litres with water and check the normality with potassium hydrogen phthalate solution – see below

Checking the normality of sodium hydroxide:

- Take 50ml of potassium hydrogen phthalate and add a few drops of phenolphthalein indicator.
- Titrate with 0.133N sodium hydroxide from a burette until the pink endpoint persists for 15-20 seconds.
- Normality = $3.42 / \text{Volume of sodium hydroxide used}$

If the normality is outside the acceptable range 0.131 – 0.135 then use a conversion factor. For e.g.: if the normality = 0.120 then the conversion factor is $0.120 / 0.133 = 0.90$.

Method A

1. Pipette 10ml of wine sample into 250ml Erlenmeyer flask containing about 40ml of water
2. Add a few drops of phenolphthalein indicator and titrate with 0.133N sodium hydroxide from a burette until the pink endpoint persists for 15-20 seconds.
3. Note the volume of sodium hydroxide used in the titration.
4. Number of sodium hydroxide used (ml) = TA as tartaric acid (g/litre)
5. Use the conversion factor here if required.

Method B – Titration with pH meter

1. Calibrate the pH meter by using each of the buffer solution of pH 3.5 and 4.
2. Take 100ml of dilutes sample in a beaker.
3. Slowly add 0.133N sodium hydroxide to the sample and titrate.
4. Place the electrodes in them, but do not allow it to touch the bottom.
5. A pH of 8.1 represents the endpoint.

A pH meter gives a more accurate value of acidity and red wines do not need to be de-colourised.

Notes:

- For red wines, samples should be de-colourised by shaking with approximately 0.5g activated charcoal and then filter through a Whatman No 1 filter paper.
- When analysing fermenting young wines, they must be first de-gassed. It can be done by bringing the sample to boil for 30 seconds. Once cooled to its original temperature it is ready to be analysed.

DETERMINATION OF VOLATILE ACIDITY

High volatile acidity indicates spoilage and effects the flavour of wine. Small amount of acetic and other volatile acids are by-products of a healthy fermentation and with red wines they form an important part of wine characteristics. The volatile acidity of wine is determined by steam distillation.

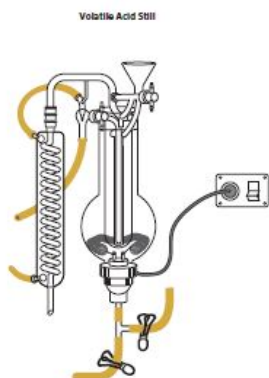
Reagents and Equipment



- 0.1N sodium hydroxide: Dissolve 0.4g sodium hydroxide pellets in 500ml water and dilute to 1 litre with water. Check the normality as described in determining total acidity.
- 10 % mercuric acid: Dissolve 1g of red mercuric oxide in 100ml of 10 % sulfuric acid. Mercuric acid is very poisonous- take great care.
- Markham Still apparatus
- Flasks
- Pipette and Burette

Method

1. Place 300ml of water and few glass beads in 1 litre boiling flask (A). Connect to the condenser and apply heat.
2. Place another 250ml flask (D) containing 10ml water under the outlet of the condenser.
3. Pipette 5ml sample into the inlet to the Markham apparatus and clipping open (e) and close (f)
4. Add 1ml of mercuric acid solution and rinse the side stopper with water. Replace the stopper and fill stopper funnel with water.
5. When the water boils open (f) and close (e). Turn on the cooling water in condenser.
6. Slowly clamp the water outlet (g) and steam until 100ml condensate is collected in the flask D.
7. Remove the flask D and open (e) and close (f). Remove the stopper and rinse the inner wall.
8. Now distil blank, repeat the previous step using 5ml distilled water instead of wine.
9. Add 5 drops of phenolphthalein indicator to the distillate flask D and titrate with 0.01N sodium hydroxide until a definite pink colour persists for 15 seconds.
10. Record the Titrate wine sample and Titrate blank
11. Calculation: Volatile acidity (g/L) = (Titrate sample – Titrate blank) x 0.12



DETERMINING PH

It is important to understand the pH of wine as it strongly influences the properties of wine such as colour, oxidation, microbiological and chemical stability. It measures the strength and concentration of the acids in relation to other compounds present in the wine. In wine making most of the pH value ranges between 3 and 4. The lower the value of pH the more acidic the sample will be.

Litmus Paper: The litmus paper strips can be used to determine the approximate pH level as a routine during cellar work. Ready to use strips can be bought with a colour chart in the packet. Different combination of three colours indicates the pH of the sample.

pH meter: The most precise pH calculation approach is using a pH meter.

Reagents

- Buffer solution, pH 3.5: Prepare a saturated solution of potassium hydrogen tartrate. Do not store for more than 7 days
- Buffer solution, pH 4.0: Dissolve 10.21g of potassium hydrogen phthalate in 500ml of hot water and dilute with water to 1 litre.

Method

1. Calibrate the pH meter by using each of the buffer solution until both give the correct reading on the scale.
2. Measure the temperature of the sample and check is it the same as that at which pH meter is set.
3. Wash the electrodes clean and dry. Pour 50 ml of the sample into a small beaker and place the electrodes in them, but do not allow it to touch the bottom.
4. Switch ON the pH meter and record the value of the sample on the digital display.
5. After recording the pH value rinse the electrodes and pat dry it. Do not leave in the sample longer than necessary.

THE YEAST INNOCULUM

Starter Cultures

It can be expensive to use large amounts of pure yeast, and so starter cultures are also used. They are usually made about a week before they are needed but they can only be made if the winery or winemaking area has predominantly the right type of yeast, like *Saccharomyces* spp. They are produced by taking a small quantity of the desired yeast and growing it under optimal conditions before adding it to the must.

The most common methods used is to crush sound berries from the best part of vineyard about a week before then other and allow the resulting must, sulphited to 100 ppm to ferment under warm conditions. This 'starter culture' is then added to the sulphited must and produces rapid fermentation eliminating the effect of undesirable micro-organisms and assure the quality of final product.

Pure Yeast Culture

When winemakers choose a cultured yeast strain, it is mostly done because the winemaker requires a predictable fermentation taken by a strain that has a reliability track record to complete.

Pure cultures can also be freeze dried and rehydrated in starter cultures. They will not only promote complete fermentation and tolerate high levels of sugar, alcohol and sulfur dioxide but also ensure the constant character and quality under cool conditions. There are many advantages of using pure yeast in active dry form such as:

- Rapid fermentation
- Better control over fermentation, thus permitting more efficient use of fermenters
- Consistable wine quality
- Reduce possibility of off flavour formation

In order to resume fermentation activity, the moisture content if dry yeast must be restored

Step 1 Rehydration

1. Using water that is between 5 and 10 times the weight of the yeast. For example, for 500 grams of dry yeast, 3 to 5 liters of rehydration water is used.
2. Rehydrate at 40-45°C water.
3. Add the yeast slowly in water to obtain even rehydration. Do not add water to yeast, allowing it to clump.
4. Allow yeast to stay in warm water for 5 to 10 minutes before stirring.
5. Yeast left in water for more than 30 minutes will reduce its activity.

Step 2 Amelioration

1. By stirring, mix the partially settled rehydrated yeast.



2. The yeast culture should ideally be cooled slowly to 5-10°C of the juice/must be inoculated; this process seems to be most important for juices/musts that are difficult to ferment.
3. Cooling can be accomplished by adding appropriate volumes of clean water or juice (with low residual SO₂) over a period of 10-20 minutes from the tank to be inoculated. Steps of 5-10°C are typical.
4. The yeast should be inoculated immediately as water alone is used after the cooling steps have been completed to prevent the cells from being inactivated. If this is not feasible, add to the yeast culture an equivalent volume of juice and inoculate within several hours and before the sugars are exhausted. However, significant foaming can result.

Step 3 Inoculation

1. When inoculated, the juice/must needs to equal or surpass 15°C to take advantage of the yeast culture over indigenous strains.
2. Active fermentation typically initiates within 24 hours. If not, by microscopic inspection of a juice/must test, verify the proportion of budding and viable yeast (Total and non-viable counts of yeast cells in a culture). If these indicators are poor, reinoculation with a freshly prepared yeast culture may be needed.
3. Once active fermentation begins, the fermentation temperature can be controlled within a small range. There have been records of temperature increases reaching 3-5 ° C per day to partially inactivate budding yeast.

